

Design Rulebook

Project Goals

Generate 3D printable electromagnetic circuits in OpenSCAD. Circuits will consist of a protopasta conductive PLA trace layer and an insulating layer that separates the traces.

Physical Laws

- Conductive trace layer resistivity (ρ): 15 $\Omega \cdot \text{cm}$.
- Resistance: $R = \frac{\rho A_c}{L}$.
- A_c = cross-sectional area of material.
- L = length of material.

Design Principles

Rules listed below are to be consulted differently for each circuit element.

Circuit elements consist of:

- DIP components
- Passive components
- Active components
- Routing
- Insulating layer

Passive Components Using Conductive Trace Geometry

- Minimum conductive trace width: 1mm.
- Minimum conductive trace spacing: 1mm.
- Conductive trace height: 1mm.
- Only refer to allowed geometry styles when constructing components
- All serpentine bending should be constructed using 45-degree chamfered segments

Routing

- Maximum route bending angle: 45 degrees.
- Vias are to be utilized to pull conductive traces out of plane when a passover is required to avoid shorting
- Via minimum height: 2mm
- Via minimum radius: 1mm
- Minimum spacing between traces: 1mm

- 4 mm by 4 mm contact pads to be placed at the power and ground ends of the circuit, with full contact between the end of the trace and the contact pad.
- Contact pads should be perfectly insulated on each side from other conductive trace elements, except the trace end they are connected to.

Insulating Layer

- Insulating layer and Conductive Trace layer are generated separately, not as a single OpenSCAD project.
- The insulating layer should provide 2mm of clearance on each trace boundary
 - i.e., the trace footprint outline is 25.4 mm by 25.4 mm; the insulating layer footprint should be 29.4 mm by 29.4 mm, matching the general shape of the trace footprint outline.

Allowed Geometry Styles for Passive Components

Resistors

- Serpentine

Capacitors

- Interdigitated

Inductors

- Planar

Optimization Goals

N/A

Style Preferences

- The insulating layer serves as a flat platform for conductive traces to be printed on.
- Only create walls in the insulating layer if necessary to isolate distinct trace planes.